METHODS / THEORY / REPRODUCIBLE EXPERIMENTS

Surface envelopes versus the separated shortcut

An accuracy audit and a paired toy-experiment study

Absolute and quantile residuals

September 9, 2026

Abstract

The surface-envelope construction tightens the old transductively standardized conformal shortcut by bounding an entire location–scale ratio, rather than separating its extrema. We recheck the construction, finite-sample proof, signed inverse, search rule, and numerical boundary conventions, and add 2,400 fresh fitted trials spanning four noise distributions, calibration sizes, and coverage targets. With identical nonnegative scores, the envelope is uniformly weakly smaller in every coordinate while preserving marginal joint validity. All new paired containment checks support that statement. Signed quantile residuals offer additional gains but change the score representation and do not have this uniform comparison. Timings also show tradeoffs: speed depends on residual type and sample size. Nine figures connect the proof to coverage, volume, coordinates, fixed-input geometry, and computation. The complete mathematical development follows in the appendix.

1 What "uniform superiority" means here

For the same nonnegative residual function, calibration observations, target, and compatible closed-cell conventions, the proved relationship is

$$C_{\text{full}}(x) \subseteq C_{\text{env}}(x) \subseteq C_{\text{old}}(x), \qquad 1-\alpha \le \Pr\{Y \in C_{\text{env}}(X)\} \le \Pr\{Y \in C_{\text{old}}(X)\}.$$

Thus superiority means *weakly smaller valid regions*. The smaller region need not cover more test observations; on paired observations its coverage cannot exceed that of its superset. Equality of some coordinates is allowed. Neither strict improvement on every sample nor universal runtime dominance is part of the result.

Audit conclusion. The core envelope argument remains valid. This revision makes the old GWC comparison explicit, handles the projected mean when GWC clips a coordinate, and separates mathematical guarantees from floating-point implementation checks and empirical findings.

In the primary six-outcome grid there are 1,440 paired fitted trials. No coordinate containment violation occurs for either absolute or capped quantile scores. All 1,440 absolute-volume pairs are finite; 1,342 capped pairs are finite. Their equal-trial mean volume reductions are 3.14% and 1.95%, respectively. The latter is conditional on a finite old volume, with 98 infinite-reference trials reported separately. These pooled numbers summarize this grid; they are not distribution-free effect-size guarantees.

2 The construction and its geometric meaning

Write m_j, s_j for the calibration mean and population-divisor standard deviation, and let $N = n + 1$. Adding a candidate coordinate z gives

$$\mu_j(z) = \frac{nm_j + z}{N}, \qquad \sigma_j(z) = \sqrt{s_j^2 + (z - m_j)^2/N}, \qquad F_j(t, z) = \frac{t - \mu_j(z)}{\sigma_j(z)}.$$

On an active cell I_j and off-coordinate GWC domains G_ℓ, the envelope uses the exact surface score

$$\max\left\{\sup_{z \in I_j} F_j(t_j, z),\ \max_{\ell \neq j} \sup_{z \in G_\ell} F_\ell(t_\ell, z)\right\}.$$

The old shortcut instead bounds $t/\sigma - \mu/\sigma$ by extrema that may occur at different candidates. For $t \geq 0$, this separation can only increase the surface upper bound. Order-statistic monotonicity and the increasing self-score inverse carry the inequality through to coordinate thresholds. The detailed proof and all endpoint conditions appear in Appendix F.

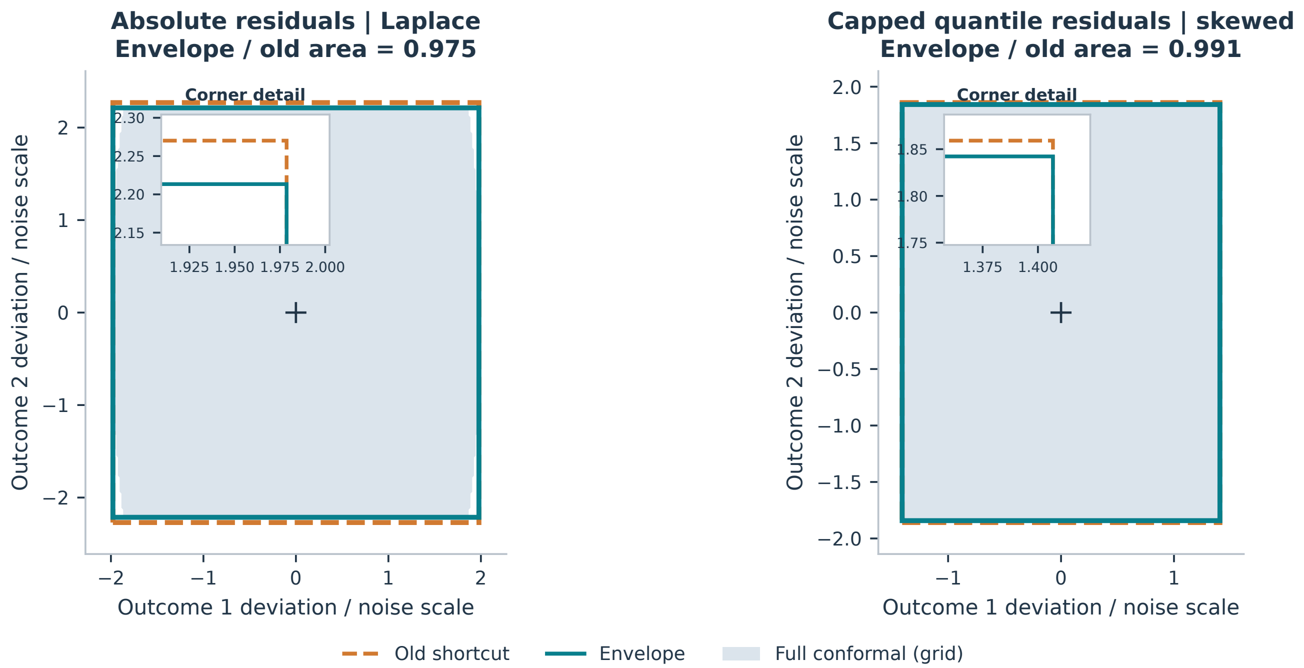

Figure 1: One input, two residual constructions. Each panel uses a fresh fitted two-outcome toy, $n = 30$, trial 0 and test input 0, fixed before plotting. The dashed old boundary encloses the envelope. Insets show the small boundary differences without enlarging them in the main axes. Gray marks full-conformal acceptance evaluated on a 241×241 outcome grid; all 85,000 accepted grid candidates lie inside the envelope. Grid shading illustrates geometry and is not an exact volume or coverage estimate. Axes are centered at the fitted point prediction (left) or quantile midpoint (right), and divided by each coordinate's known DGP noise scale.

The envelope encloses accepted coordinate pieces; it can fill gaps and need not equal the full-conformal set. Here the respective area ratios are 0.975 and 0.991. The examples are deliberately fixed, so they also convey that a valid gain can be visually modest. The endpoint scale normalization is only for display and leaves area ratios unchanged.

3 Experimental design and estimands

Every trial generates independent training, calibration, and test observations and refits the models. Methods within a trial share all observations and fitted predictions. There is no reused simulation pool or fitted model across trials. We use 800 training and 1,200 test observations, three Gaussian features, and six outcomes unless specified otherwise.

Let $g(x) = \mathbf{1}\{x_1 > 0\}$, a_j be geometrically spaced from 0.6 to 2.4, and $B_{kj} = \frac{1}{2}\sin(\ell_{kj} + 1)$, where ℓ_{kj} is the zero-based row-major index in a $3 \times d$ matrix. The data-generating model is

$$X \sim N(0, I_3), \qquad Y_j = X^\top B_{\cdot j} + (1 + 0.8g(X))a_j\epsilon_j.$$

The four noise laws are independent standard Gaussian; Gaussian with equicorrelation 0.8; independent unit-variance Laplace; and independent centered, unit-variance $\exp(0.8Z)$ with $Z \sim N(0, 1)$. Thus every family includes covariate-dependent scale and heterogeneous outcome units, and the last adds skewness.

Fitted absolute and quantile scores. An OLS model with intercept supplies $\hat{f}_j$. For each group and outcome, the empirical training-error quantiles at $\beta/2$ and $1 - \beta/2$ supply offsets to $\hat{f}_j$, yielding ordered $\hat{q}_j^-, \hat{q}_j^+$. This is a simple fitted location model with groupwise empirical quantile offsets, not an oracle quantile model or a gradient-boosted quantile regressor. All offsets are estimated from training data alone. With $w_j = \hat{q}_j^+ - \hat{q}_j^-$:

$$E_j^{\text{abs}} = |Y_j - \hat{f}_j(X)|, \qquad E_j^{\text{raw}} = \max(\hat{q}_j^-(X) - Y_j, Y_j - \hat{q}_j^+(X)), \qquad E_j^{\text{cap}} = \max(E_j^{\text{raw}}, 0).$$

The raw score and possible contraction of a fitted base interval follow the CQR construction of Romano et al. [2]. Here their coordinate scores are used inside the audited multivariate envelope. Raw scores have test-specific lower endpoint $-w_j(x)/2$. The old shortcut is evaluated on absolute or capped scores; raw signed scores are not fed into its nonnegative-score formula.

Prespecified grids. The primary grid crosses four noise laws with $n \in \{30, 80, 200\}$ at $\alpha = \beta = 0.1$, with 120 trials per cell (1,440 trials). At $n = 80$, Gaussian and Laplace toys additionally use $\alpha \in \{0.05, 0.2, 0.3\}$ (720 trials). Two Gaussian, two-outcome studies at $n = 80$ use $\beta \in \{0.1, 0.02\}$ and $\alpha = 0.1$ (240 trials). Two further fitted datasets supply Figure 1; they are not included in the Monte Carlo summaries.

Coverage, size, and uncertainty. Joint coverage is the within-trial proportion covering all outcomes. Lengths are $2L_j$ for absolute scores and $w_j(x) + 2L_j$ for quantile scores; volumes are products of outcome lengths, averaged over the trial's test inputs. Empty boxes have zero volume. We plot the mean of the *paired trial ratios* $V_{\text{env},b}/V_{\text{old},b}$, not a ratio of pooled mean volumes. Ratios use finite positive denominators and finite numerators; excluded counts are shown. Coverage includes every trial, including infinite regions. Unless stated otherwise, error bars are 95% pointwise Monte Carlo intervals $\bar{z} \pm t_{R-1,0.975}s_z/\sqrt{R}$, computed across independent fitted trials (or the reported finite subset), not across pooled test observations. They are approximate intervals, not simultaneous inference or a validity proof.

4 Absolute residuals: gains concentrate at small calibration sizes

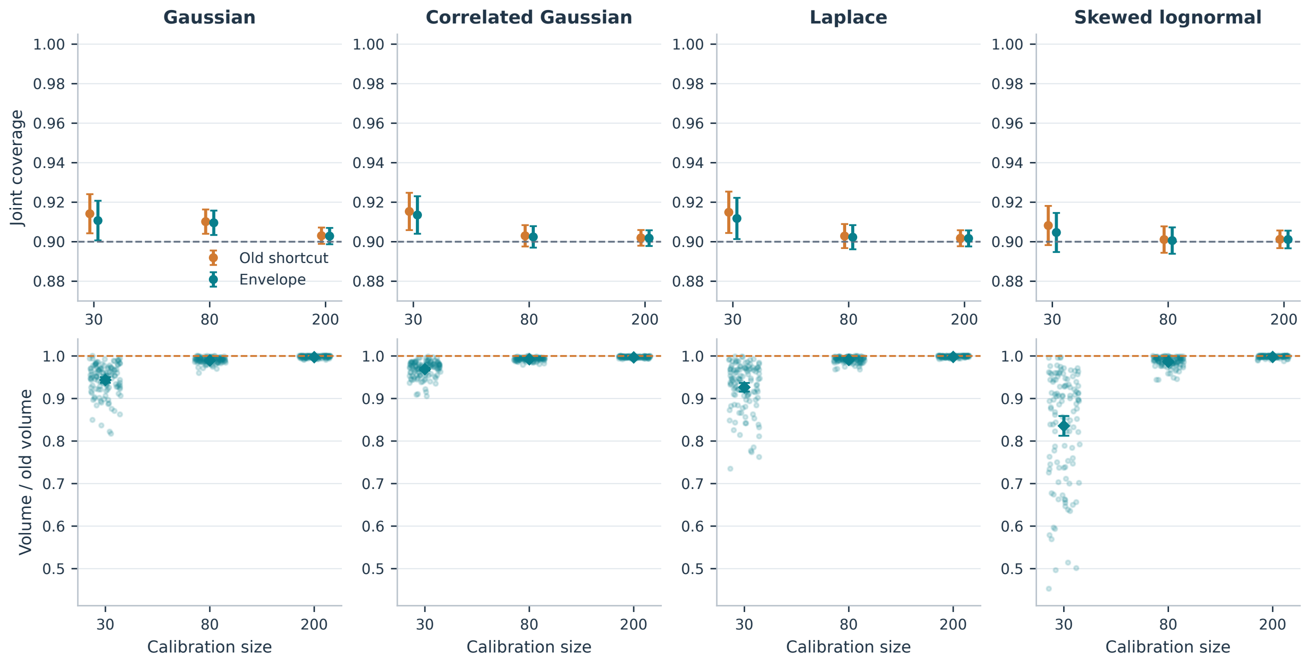

Figure 2: Valid size reduction across four noise laws. Six outcomes, 120 independent fitted trials per panel location, target 0.9. Top: mean joint coverage with 95% Monte Carlo intervals. Bottom: all paired outcome-volume ratios (faint dots), their means (diamonds), and intervals. The dashed ratio line marks equality. The shared vertical scale prevents small gains in one family from appearing larger than substantial gains in another. All 1,440 volume pairs are finite.

Across the four distributions, the average paired volume reduction is 8.13% at $n = 30$, 1.03% at $n = 80$, and 0.27% at $n = 200$. Gains diminish as the candidate contributes less to augmented calibration statistics. This is an observed trend on this grid, not a proven monotonicity in n. Skewed noise produces the largest small-sample reductions and appreciable between-trial variation. Every observed ratio remains below one.

Scores	Noise	Old cov.	Env. cov.	Reduction	Pairs
Absolute	Gaussian	0.9102	0.9096	$1.05 \pm 0.12\%$	120
Capped CQR	Gaussian	0.9062	0.9060	$0.22 \pm 0.04\%$	120
Absolute	Correlated Gaussian	0.9030	0.9024	$0.77 \pm 0.07\%$	120
Capped CQR	Correlated Gaussian	0.9052	0.9050	$0.19 \pm 0.03\%$	120
Absolute	Laplace	0.9029	0.9023	$0.92 \pm 0.12\%$	120
Capped CQR	Laplace	0.9047	0.9045	$0.33 \pm 0.05\%$	120
Absolute	Skewed lognormal	0.9011	0.9006	$1.37 \pm 0.22\%$	120
Capped CQR	Skewed lognormal	0.8997	0.8995	$0.46 \pm 0.07\%$	120

Table 1: A common calibration-size comparison. Six outcomes, $n = 80$, target 0.9. Reduction is $100(1 - \overline{V_{\text{env}}/V_{\text{old}}})$; the $\pm$ term is its 95% Monte Carlo interval half-width. Pairs count finite-volume comparisons. Capped CQR is included for direct comparison with the next section. A point estimate just below 0.9 is possible under Monte Carlo variation; the guarantee concerns the population experiment.

5 Capped quantile residuals: the same theorem, additional edge cases

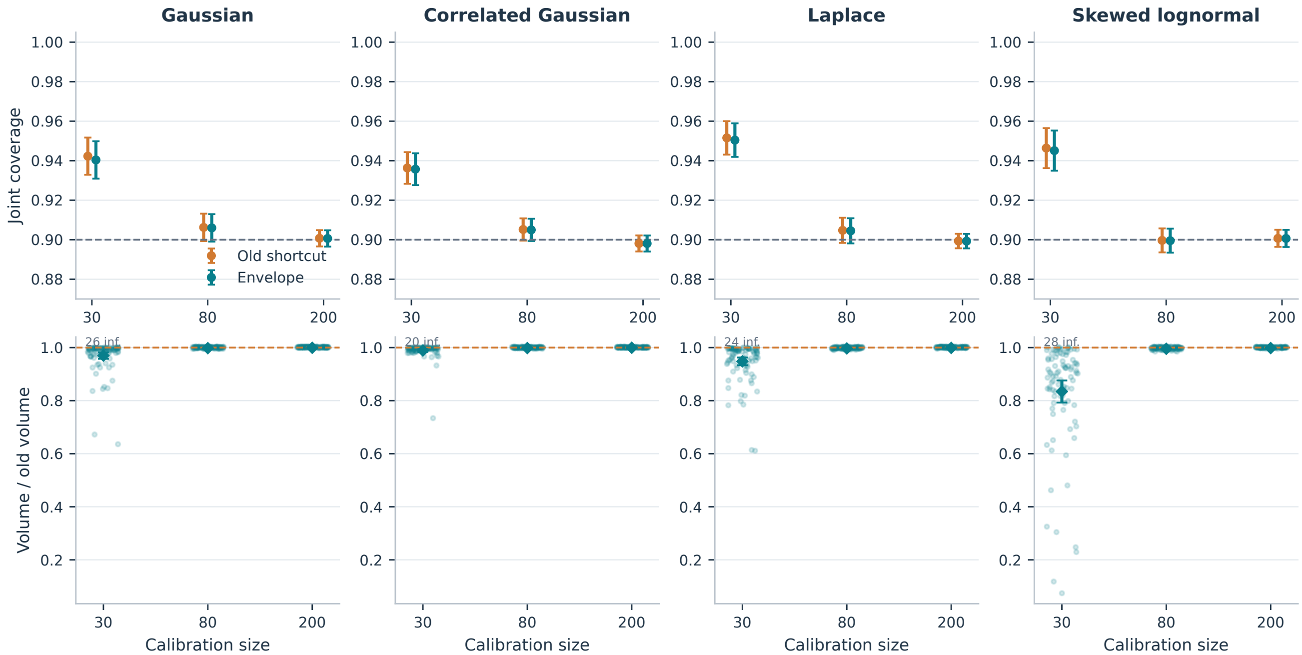

Figure 3: Capped CQR isolates the envelope refinement. Both methods receive exactly the same capped calibration residuals and fitted base intervals. Coverage includes infinite-region trials; bottom-row ratios are restricted to finite pairs. The labels "inf." count excluded pairs with an infinite old volume. The envelope itself may be finite in such a pair, so the finite-pair gain omits that improvement. All axes and uncertainty conventions match Figure 2.

At $n = 80$ the mean reductions range from 0.19% to 0.46% (Table 1). These are consistent gains from the construction alone, and they are considerably smaller than many signed-versus-capped gains. Keeping the residual representation fixed makes the interpretation precise.

At $n = 30$, the old shortcut has infinite volume in 98 of 480 trials (20.4%); the capped envelope does so in 88 (18.3%). In 87 of these small-sample trials, at least one capped coordinate has zero calibration variance. Both methods then use the same explicitly documented whole-domain fallback. Additional infinities can arise when the linked threshold reaches the finite range limit of the self-score. No infinite trial is replaced by a large plotting constant. Neither method has an infinite volume at $n = 80$ or $n = 200$ in this primary grid.

Read size and coverage together. Capping creates a mass at zero. Small-sample fallback can make coverage conservative and mean volume infinite. Report the infinity rate next to finite-pair efficiency; the latter does not estimate unconditional expected volume.

6 Uniformity and coordinate-by-coordinate comparisons

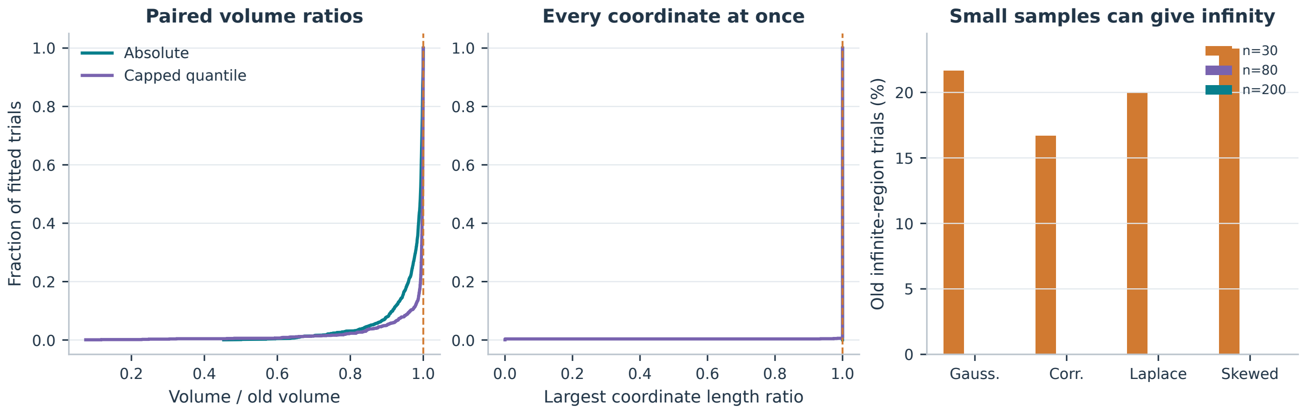

Figure 4: The full paired distribution, not just its mean. The first two panels pool the primary grid: 1,440 finite absolute pairs and 1,342 finite capped pairs. For each trial, the second panel takes the *largest* of the six coordinate length ratios, so a value above one would expose a coordinate violation. Many coordinates coincide to numerical precision. The final panel reports old-method infinity frequencies rather than hiding these trials in the ratio distribution. A finite empirical audit supplements the pathwise proof; it does not replace it.

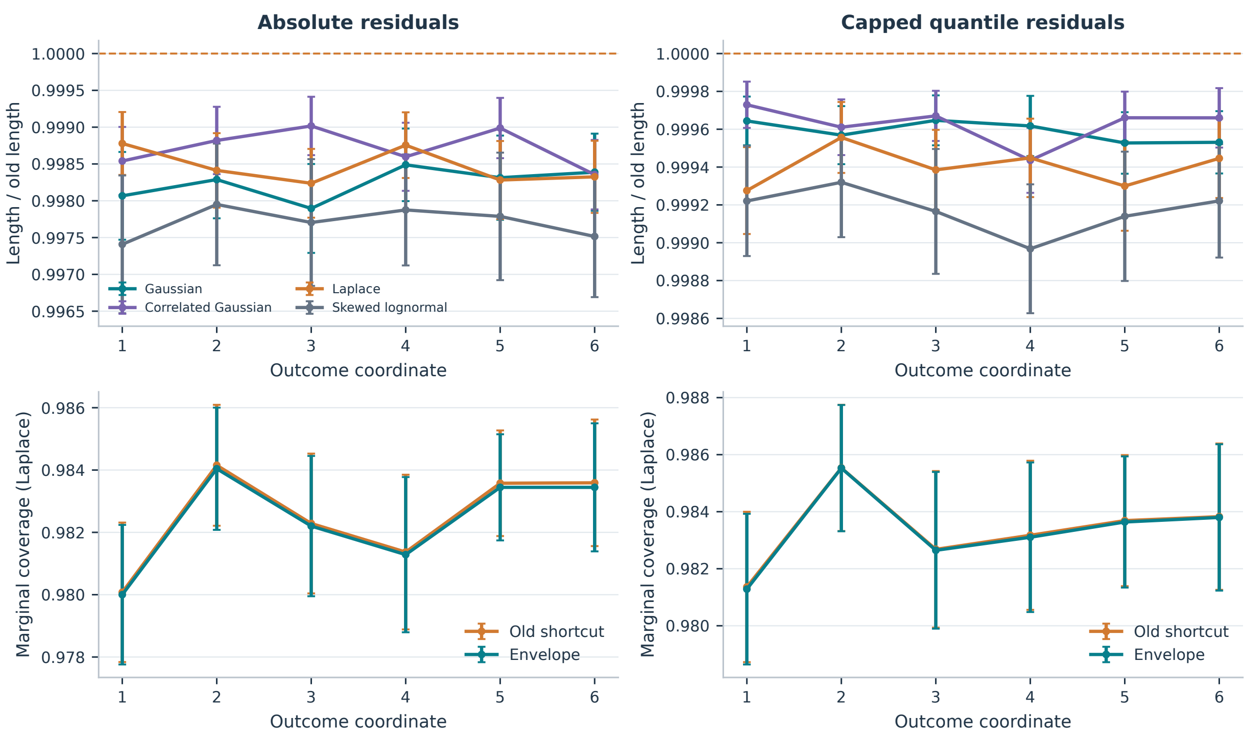

Figure 5: Shorter in each coordinate, with coverage still visible. Top: paired mean length ratios and 95% Monte Carlo intervals for all four families at $n = 80$, $d = 6$. Bottom: marginal coverages for the Laplace study, computed on the same trials. A high marginal coverage is not a substitute for joint coverage; the latter is reported separately. Normalization by the old length removes arbitrary outcome-unit differences.

The largest finite volume ratios are 0.999968 (absolute) and 0.999992 (capped). Maximum coordinate ratios are within 2×10^{-13} of one; these near-equalities are not practically meaningful strict gains. Both same-score comparisons pass on all 2,400 fitted trials, including common fallback cases.

7 Coverage targets and volumes in outcome units

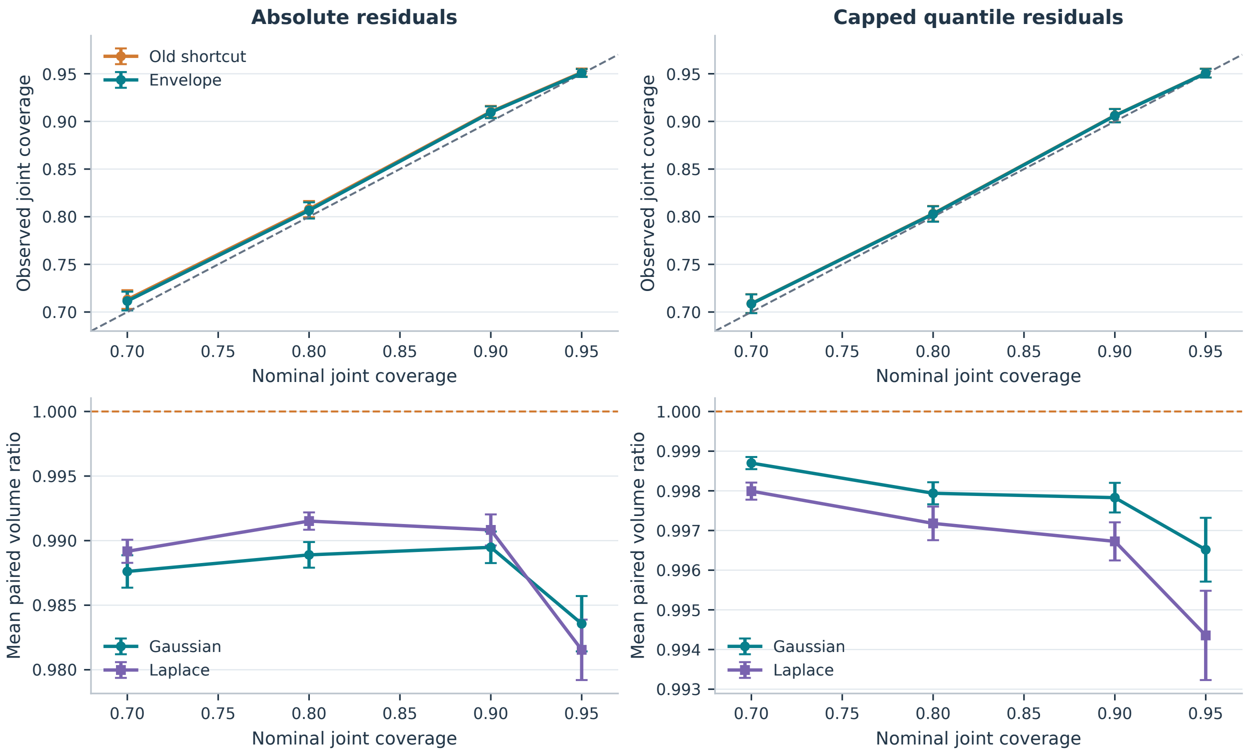

Figure 6: Changing the target does not change the comparison's meaning. At $n = 80$, $d = 6$, base miscoverage stays fixed at 0.1 while the simultaneous target varies. Top: Gaussian joint coverage versus nominal coverage, with the identity line. Bottom: paired volume ratios for Gaussian and Laplace noise. The four target values use 120 independently fitted trials each; the 0.9-target trials are the primary-grid data already shown, not extra replications. Error bars are pointwise 95% Monte Carlo intervals.

Scores	Noise	Old mean volume	Envelope mean volume
Absolute	Gaussian	796235.2	787770.3
Capped CQR	Gaussian	613944.2	612547.1
Absolute	Correlated Gaussian	287799.9	285667.6
Capped CQR	Correlated Gaussian	293560.1	292957.4
Absolute	Laplace	2012805.1	1993559.9
Capped CQR	Laplace	2093510.5	2085515.1
Absolute	Skewed lognormal	3924120.3	3869557.1
Capped CQR	Skewed lognormal	2099168.0	2087665.9

Table 2: Volumes on the original outcome scale. Arithmetic mean six-dimensional volumes at $n = 80$, corresponding to Table 1. All pairs here are finite. Multiplying the six outcome lengths gives large numbers even when coordinate changes are small. The ratio of these displayed means generally differs from the mean paired ratio used in the plots.

Joint nominal coverage and base marginal coverage are different quantities. For independent outcomes, even exactly fitted 90% marginal intervals would have joint coverage $0.9^6 = 0.5314$. This calculation explains why calibration often expands the base intervals; the fitted toy's actual base coverage is measured from data and need not equal this idealized value.

8 Signed quantile residuals: separate the sources of improvement

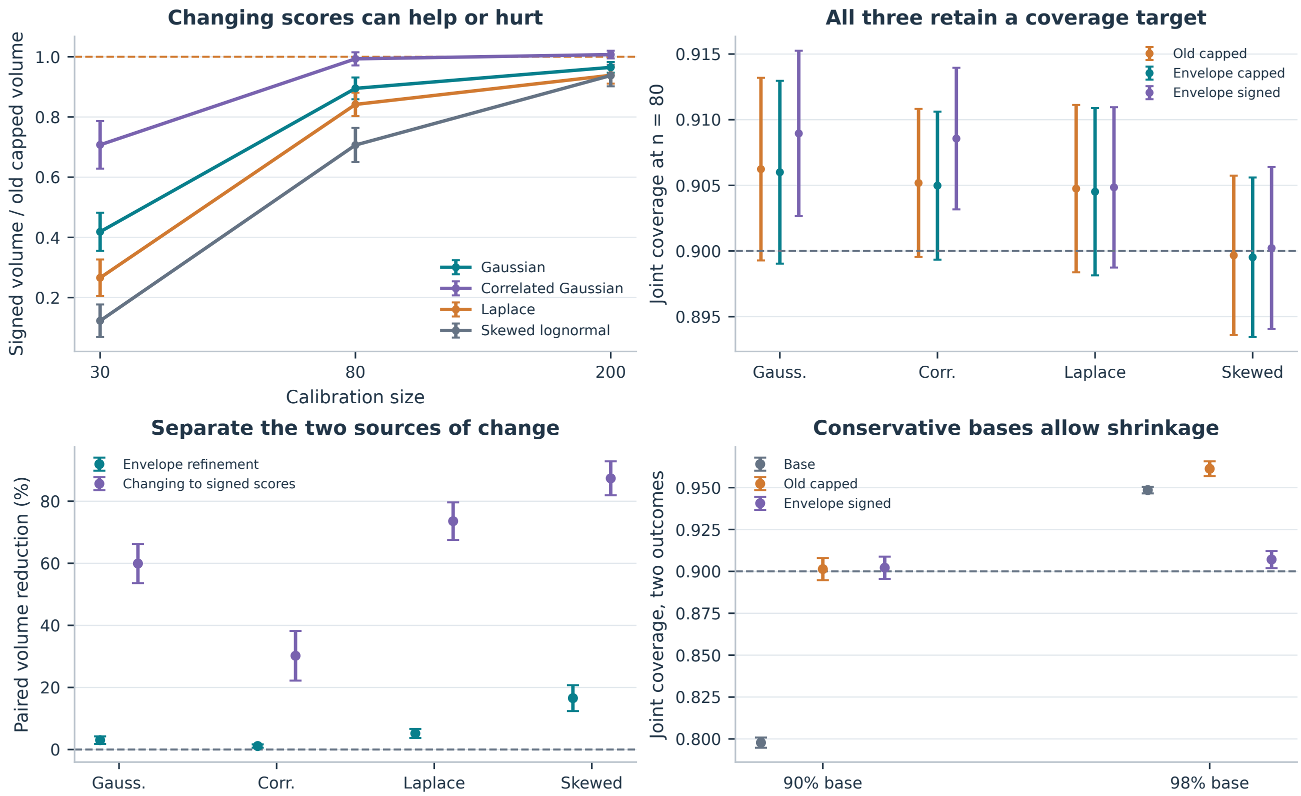

Figure 7: A score change is not the same as an envelope refinement. Top left: signed-envelope / old-capped paired volume ratios on finite pairs; crossing one means the signed version is larger. Top right: all-trial joint coverage at $n = 80$. Bottom left: at $n = 30$, the two comparisons are envelope-capped / old-capped and envelope-signed / envelope-capped, each using its own finite-pair denominator set. Their reductions are not additive. Bottom right: two-outcome Gaussian toys with 90% and 98% fitted marginal bases, both calibrated to a 90% joint target. All error bars are 95% Monte Carlo intervals across fitted trials.

The signed envelope is smaller on average than the old capped shortcut in many configurations: at $n = 80$, the finite-pair mean reductions are 10.49% (Gaussian), 0.68% (correlated Gaussian), 15.83% (Laplace), and 29.32% (skewed lognormal). Much of this change comes from retaining signed residuals, not from tightening the same capped construction. The particularly large small-n gains are also sensitive to capped-score scale degeneracy and heavy-tailed finite volumes; the finite-pair conditioning must be retained.

There is no samplewise signed-versus-capped superiority: the signed envelope has larger volume in 359 of the 1,342 finite primary-grid comparisons, and the largest observed ratio is 2.081. For correlated Gaussian noise at $n = 200$, its mean ratio is 1.007, with 95% interval $[0.995, 1.020]$. This mean is statistically compatible with equality; the individual larger regions already rule out uniform samplewise containment.

What is proved for signed scores? $C_{\text{full,raw}} \subseteq C_{\text{env,raw}} \subseteq C_{\text{GWC,raw}}$ and marginal joint coverage at least the target. The comparison to a capped old shortcut is empirical and changes the full-conformal reference set.

9 One test input: expansion, contraction, and a concrete exception

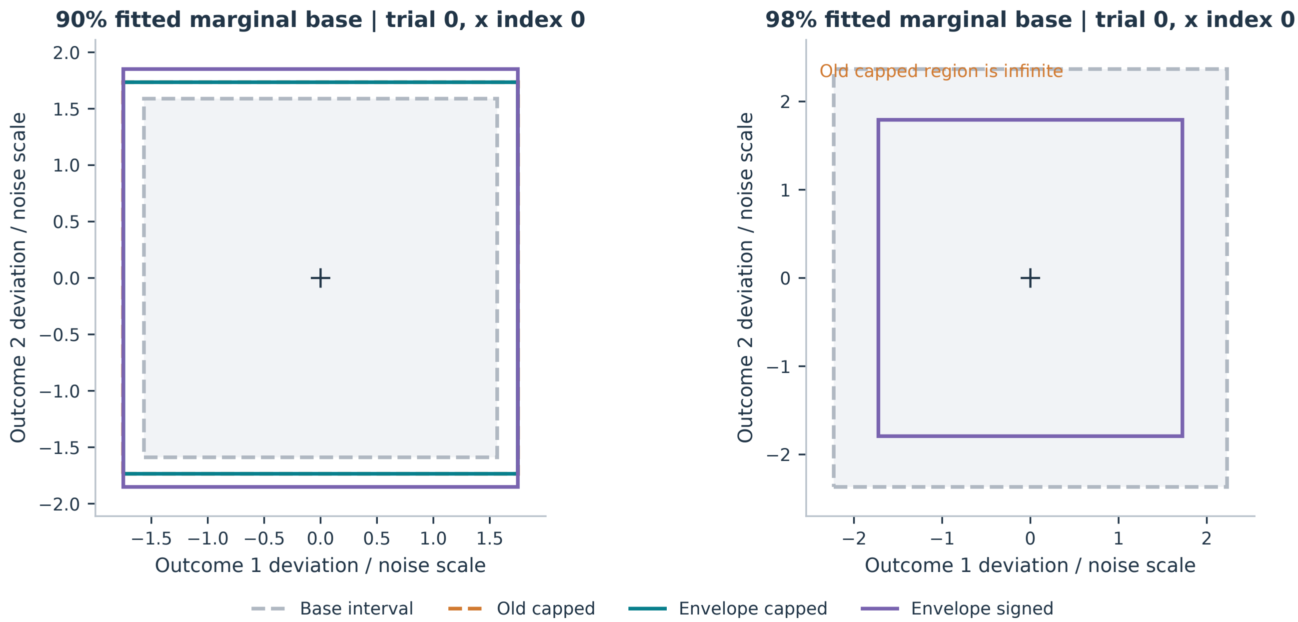

Figure 8: Fixed-input signed geometry, without selecting a favorable trial. Both panels use trial 0 and test input 0 from the prespecified two-outcome Gaussian studies. Coordinates are centered at the fitted quantile midpoint and normalized by the DGP coordinate scales. Left: a 90% base must expand, and the signed envelope extends farther along outcome 2 than the capped construction. Right: a conservative 98% base contracts under signed calibration; both capped methods are infinite on this trial. The unbounded regions are stated explicitly and are not drawn as finite boxes.

In the left panel, normalized outcome-2 half-widths are 1.735 for the old capped shortcut and 1.851 for the signed envelope. This fixed example makes the absence of signed-versus-capped containment tangible. In the right panel, the base half-widths $(2.230, 2.368)$ contract to $(1.723, 1.792)$.

Across all 120 conservative-base trials, the signed envelope has joint coverage 0.9071 and mean outcome area 52.37, versus base coverage 0.9485 and area 62.53. The ratio of these mean areas implies a 16.25% reduction relative to the uncalibrated base; it is *not* a paired reduction relative to the old shortcut. Both capped methods have infinite volume in 29/120 trials. Signed thresholds are negative in 88.75% of test-coordinate evaluations, averaged over trials. The 90%-base study instead has signed coverage 0.9022 and needs expansion in most coordinates.

A useful invariance check. For a fixed coordinate shift h_j, shifting every calibration score and the test-domain endpoint by $-h_j$ shifts the exact envelope threshold by $-h_j$. Indeed m and z shift together, s is unchanged, and F is unchanged. Thus with a covariate-independent symmetric base of half-width h, raw scores $|Y - \hat{f}| - h$ reproduce the absolute-envelope outcome intervals, up to numerical roundoff. A constant shift alone cannot create an efficiency gain. Our groupwise bases vary with the covariate, and capping is nonlinear, so this invariance does not imply equality in the reported signed-versus-capped study.

10 Computation: measured speed and its limits

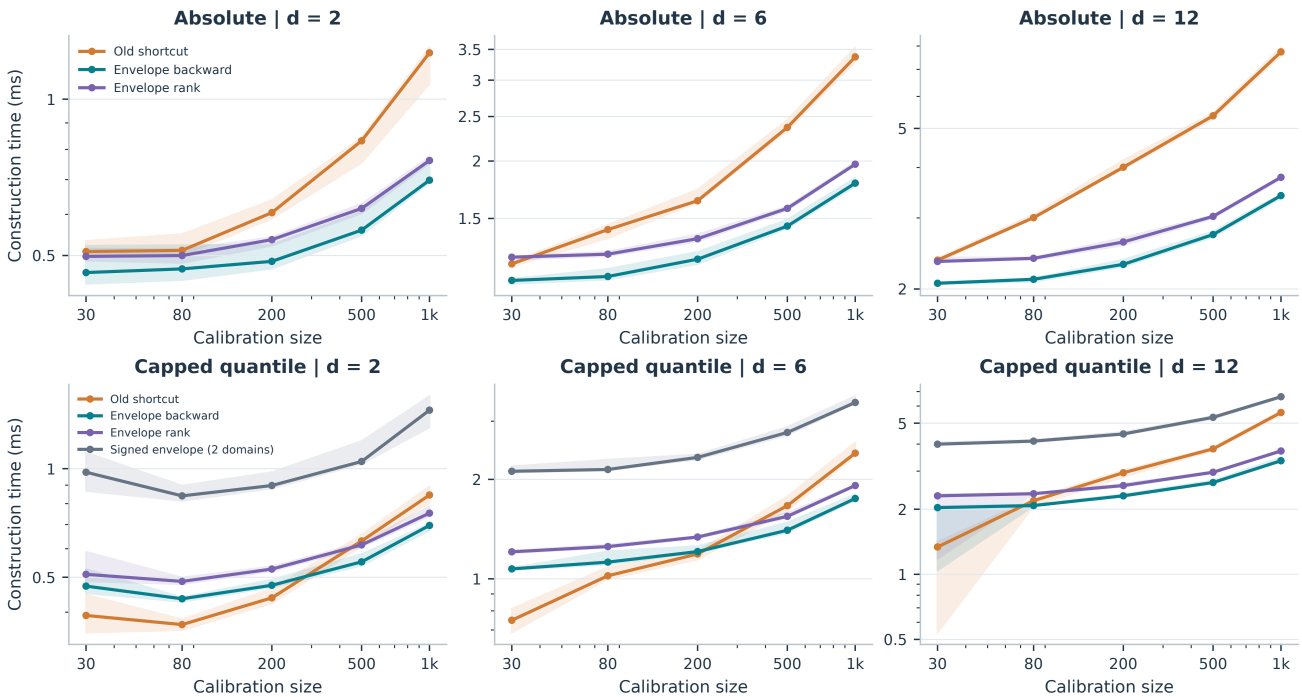

Figure 9: Serial region-construction timings. Each (n, d) cell uses 15 newly fitted Gaussian datasets. For each method, one untimed warmup is followed by three timed calls in randomized method order; the dataset-level statistic is their median. Lines show the median across datasets and bands the interquartile range, not confidence intervals. Both axes are logarithmic; panels use individual time ranges. Timing includes statistics, sorting, and search, and excludes model fitting, evaluation, plotting, and file I/O. Capped panels add raw signed envelope timing for both distinct test-width domains using one calibration object. These are two predictions, whereas a capped region is computed once for all test inputs.

The absolute-envelope backward implementation is faster than the old shortcut at all 15 plotted (n, d) settings in this run; the ratios of median old time to median envelope time range from 1.09 to 2.27. Capped comparisons range from 0.66 to 1.67, so several small-sample cases favor the old shortcut. Zero-scale fallbacks make some small capped cases particularly cheap. These timings describe the supplied Python implementations on this Windows host, not an algorithmic guarantee of uniform speed or a hardware-independent benchmark.

The envelope precomputes off-coordinate maxima in $O(nd)$ and uses $O(n)$ work per surface quantile. Unconditional backward search has worst-case $O(dn^2 + dn \log n)$ cost. The optional rank localization prunes a provably failing suffix and returns identical bounds in the benchmark checks. Its extra order statistic can cost more than it saves when backward search already visits only one cell. Neither an unproved global binary-search predicate nor full multidimensional cell enumeration is used here. The exact search proofs and their qualified complexity bounds are retained in the appendix.

11 Accuracy audit, reproduction, and scope

Mathematical checks. The augmented scale uses the sample divisor n on $n+1$ augmented observations; the calibration s_j^2 uses divisor n on n observations. The signed inverse keeps the sign of its argument. Coordinate suprema include endpoints and only an admissible interior *maximum*; negative suprema are not clamped to zero. The conformal rank is $\lceil (n+1)(1-\alpha) \rceil$, with $+\infty$ when it exceeds n. Candidate acceptance and cell boundaries are weak/closed. Thresholds below the attainable domain mean an empty set, not a singleton.

The old global rule agrees algebraically with the exact GWC score for nonnegative residuals before its positive jitter. The revised proof states this connection explicitly. It also uses the minimum scale on the clipped GWC interval when that interval does not contain the mean, rather than silently substituting the unrestricted calibration scale. The code comparison retains the old implementation's corrected tied-mean indexing and 10^{-10} positive jitter. The report does not equate that floating-point executable to the closed-cell mathematical comparator for arbitrary ties. New numerical comparisons use tolerance $10^{-8}(1+|L_j^{\text{old}}|)$.

Executed verification. The formula audit passed 150 augmented-standardization checks, 150 inverse checks, 1,200 interval optimizations, 400 backward/exhaustive/rank agreements, and 600 nonnegative comparisons. Of 33,651 candidate checks, all 10,683 full conformal acceptances were retained. The protocol regression passed fresh-fit checks, 500 atom-heavy draws (positive-scale cases checked), and 900 exact-zero boundary checks. The new suite adds 2,400 pairs per same-score comparison, 450 benchmark search agreements, and the two geometry grids. These finite checks support the supplied derivations, not a claim of exhaustive software verification. All 2,400 archive hashes and distinct training-data hashes were checked, and every saved trial metric was recomputed. Full method bounds were also recomputed on 60 prespecified archived trials; 120 constant-shift invariance checks passed.

Reproducible outputs. The directory `envelope_method/report_revision/` contains `design.json`, the deterministic generator and fitting code, `trials.csv`, coordinate metrics, containment records, runtime observations, summary tables, and figure code. Each of the 2,400 fitted trial archives stores training/calibration/test features and outcomes, fitted coefficients, quantile offsets, predictions, residuals, domains, and method bounds. The manifest records SHA-256 digests. The master seed is 2026090917; configuration k, trial b uses seed $2026090917 + 10000k + b$. Auxiliary benchmark and geometry seeds are specified in the same runner. No historical synthetic results are reused in the reported Monte Carlo summaries.

The README gives the exact simulation, figure, assembly, compilation, environment, and rendering commands. Runtime reruns vary with system load. Formula checks and the figure-generation grid are separate from fitted-trial coverage experiments. No new full-LWC experiment is launched.

Scope of the evidence. This is a deliberately interpretable toy study, not a comprehensive model or real-data benchmark. It tests four fixed DGP families and one lightweight quantile-fitting scheme. Guarantees are marginal over calibration and a new test observation, conditional on independent training; they are not conditional coverage guarantees at every input. The report supports uniform *weak size improvement on identical nonnegative scores*; signed-score choices and computational speed require the empirical qualifications above.

A Setup and the conformal rank

Condition throughout on an independently trained predictor and any training-only choices. Let $(X^i, Y^i)_{i=1}^{n+1}$ be exchangeable and let

$$\mathbf{E}^i = \big(E_1(X^i, Y_1^i), \ldots, E_d(X^i, Y_d^i)\big) \in \mathbb{R}^d$$

be coordinatewise nonconformity scores. The first n residual vectors are observed. Fix a test input x and let $\mathbf{e} = \mathbf{E}(x, y)$ denote a candidate residual. Suppose its attainable coordinate values lie in

$$D_j(x) = [a_j(x), \infty) \cap \mathbb{R}, \qquad D(x) = \prod_{j=1}^{d} D_j(x), \qquad a_j(x) \in \mathbb{R} \cup \{-\infty\}.$$

Here $a_j(x)$ is a known bound for the test residual, not an estimated bound on the distribution. Calibration residuals at other covariates need not exceed $a_j(x)$. A larger enclosing domain, including $D_j = \mathbb{R}$, is also valid. Coordinatewise score sublevel sets must be intervals for the final prediction set in outcome space to be a rectangle; validity itself does not require that.

Write $N = n + 1$ and

$$r = \lceil N(1-\alpha) \rceil, \qquad \mathcal{Q}_\alpha(v_1, \ldots, v_n) = \begin{cases} v_{(r)}, & r \le n, \\ +\infty, & r = N. \end{cases} \tag{1}$$

Use weak acceptance inequalities throughout. No jitter is needed, and ties are allowed. If $v_i \le w_i$ for every i, then $\mathcal{Q}_\alpha(v) \le \mathcal{Q}_\alpha(w)$: for each real c, at least as many v_i as w_i are at most c, so the ordered values have the same ordering. This elementary monotonicity is used repeatedly below.

Lemma A.1 (Exchangeable rank)**.** *If $S^1, \ldots, S^N$ are exchangeable real scores, then $\Pr\{S^N \le \mathcal{Q}_\alpha(S^1, \ldots, S^n)\} \ge 1 - \alpha$.*

Proof. Randomly break ties using independent continuous auxiliary variables. The rank of the last score among the N scores is uniform on $\{1, \ldots, N\}$ by exchangeability. A rank at most r implies the weak acceptance event in the statement, including when tied values occur. Its probability is therefore at least $r/N \ge 1 - \alpha$. If $r = N$, acceptance is automatic by (1). □

This is the standard conformal rank argument; see [1]. The derivations of the envelope and its comparisons below are supplied here.

B Augmented standardization and its self-score link

For each coordinate define

$$m_j = \frac{1}{n} \sum_{i=1}^{n} E_j^i, \qquad s_j^2 = \frac{1}{n} \sum_{i=1}^{n} (E_j^i - m_j)^2. \tag{2}$$

The divisor in s_j^2 is n, whereas the variance of the augmented N-point sample below has divisor $N - 1 = n$. This distinction fixes all constants in the link. Initially assume $s_j > 0$ for every j; Section H supplies a valid extension to degenerate calibration samples. Finite population moments are unnecessary for the finite-sample argument.

For a candidate coordinate z, put $u = z - m_j$. The augmented mean is

$$\mu_j(z) = \frac{n m_j + z}{N} = m_j + \frac{u}{N}. \tag{3}$$

Using $\sum_i (E_j^i - m_j) = 0$ gives the full variance expansion

$$\sum_{i=1}^{n} (E_j^i - \mu_j(z))^2 + (z - \mu_j(z))^2 = \sum_{i=1}^{n} \left(E_j^i - m_j - \frac{u}{N}\right)^2 + \left(\frac{nu}{N}\right)^2$$
$$= ns_j^2 + \frac{nu^2}{N^2} + \frac{n^2u^2}{N^2} = ns_j^2 + \frac{nu^2}{N}.$$

Consequently

$$\sigma_j(z) = \sqrt{s_j^2 + \frac{(z - m_j)^2}{N}}. \tag{4}$$

The basic coordinate transformation, its scalarization, and its candidate self-score are respectively

$$F_j(t, z) = \frac{t - \mu_j(z)}{\sigma_j(z)} = \frac{t - m_j - (z - m_j)/N}{\sqrt{s_j^2 + (z - m_j)^2/N}}, \tag{5}$$

$$\Phi_{\mathbf{z}}(\mathbf{t}) = \max_{j \le d} F_j(t_j, z_j), \tag{6}$$

$$g_j(e) = F_j(e, e) = \frac{n(e - m_j)/N}{\sqrt{s_j^2 + (e - m_j)^2/N}}. \tag{7}$$

Unlike a calibration-only fitted standardization, at the true test residual these parameters are symmetric functions of all N residual vectors.

Lemma B.1 (Signed inverse)**.** *Let $T = n/\sqrt{N}$. The map $g_j : \mathbb{R} \to (-T, T)$ is strictly increasing, and for finite e,*

$$g_j(e) \le c \quad \Longleftrightarrow \quad e \le \omega_j(c), \tag{8}$$

where extended-real endpoints encode empty and unrestricted sublevels, and

$$\omega_j(c) = \begin{cases} -\infty, & c \le -T, \\ m_j + \dfrac{N s_j c}{\sqrt{n^2 - Nc^2}}, & -T < c < T, \\ +\infty, & c \ge T. \end{cases} \tag{9}$$

Proof. Direct differentiation gives $g_j'(e) = (n/N)s_j^2\{s_j^2 + (e - m_j)^2/N\}^{-3/2} > 0$. Its limits at $\pm\infty$ are $\pm T$, never attained at finite e. For $|c| < T$, the equation $g_j(e) = c$, with $u = e - m_j$, yields

$$c\sqrt{s_j^2 + u^2/N} = nu/N, \quad (n^2 - Nc^2)u^2 = N^2 s_j^2 c^2, \quad \operatorname{sign}(u) = \operatorname{sign}(c).$$

Thus $u = N s_j c/\sqrt{n^2 - Nc^2}$. Keeping the sign condition is essential: squaring alone would introduce a spurious root for negative c. $\square$

The signed residual sublevel is $D_j(x) \cap (-\infty, \omega_j(c)]$. There is no $\max\{0, \omega_j(c)\}$ in the signed link. If the upper endpoint is below a finite $a_j(x)$, the sublevel is empty.

C The full conformal reference set

For a candidate $\mathbf{e}$ define

$$Q_{\mathbf{e}} = \mathcal{Q}_\alpha(\Phi_{\mathbf{e}}(\mathbf{E}^1), \dots, \Phi_{\mathbf{e}}(\mathbf{E}^n)), \qquad C_{\text{full}}(x) = \{y : \Phi_{\mathbf{E}(x,y)}(\mathbf{E}(x, y)) \le Q_{\mathbf{E}(x,y)}\}. \tag{10}$$

Equivalently, y is accepted if $e_j \le \omega_j(Q_{\mathbf{e}})$ for every j. The threshold depends on the entire candidate, so this reference set need not be rectangular or cheap to compute.

Theorem C.1 (Full conformal coverage). *Under exchangeability,* $\Pr\{Y^{n+1} \in C_{\text{full}}(X^{n+1})\} \geq 1-\alpha$.

Proof. At $\mathbf{e} = \mathbf{E}^{n+1}$, augmenting the calibration sample gives precisely the full exchangeable sample. The means and sample standard deviations of that sample are invariant to permutation. Applying the same standardized maximum to each of the N observations therefore gives exchangeable scalar scores. Lemma A.1 proves the assertion. One need not assume that the scores are independent of the fitted transformation. □

D Exact coordinate suprema and signed GWC

For any nonempty interval $I = [\ell, u] \cap \mathbb{R}$, allowing infinite endpoints, define

$$\psi_{j,I}(t) = \sup_{z \in I} F_j(t, z). \tag{11}$$

Endpoint limits are included when the endpoint is infinite. Suppressing j, differentiation of (5) gives

$$\frac{\partial F(t,z)}{\partial z} = -\frac{s^2 + (t-m)(z-m)}{N\{s^2 + (z-m)^2/N\}^{3/2}}. \tag{12}$$

If $t > m$, the unique critical point is a maximum,

$$z^*(t) = m - \frac{s^2}{t-m}, \qquad F(t, z^*(t)) = \sqrt{\frac{(t-m)^2}{s^2} + \frac{1}{N}}. \tag{13}$$

If $t < m$, its critical point is a minimum; if $t = m$, the function is decreasing. Also $F(t, +\infty) = -1/\sqrt{N}$ and $F(t, -\infty) = 1/\sqrt{N}$. Thus the complete closed form is

$$\psi_{j,I}(t) = \max\left\{ F_j(t,\ell), F_j(t,u), \sqrt{\frac{(t-m_j)^2}{s_j^2} + \frac{1}{N}} \;\middle|\; t > m_j,\ z^*(t) \in I \right\}. \tag{14}$$

The conditional entry is omitted when its condition fails, rather than replaced by zero. Replacing it by zero would incorrectly inflate negative suprema.

The signed GWC score, quantile, and bounds are

$$\Phi_G(\mathbf{t}; x) = \max_j \psi_{j, D_j(x)}(t_j) = \sup_{\mathbf{z} \in D(x)} \Phi_{\mathbf{z}}(\mathbf{t}), \tag{15}$$

$$Q_G(x) = \mathcal{Q}_\alpha(\Phi_G(\mathbf{E}^1; x), \ldots, \Phi_G(\mathbf{E}^n; x)), \tag{16}$$

$$U_j(x) = \omega_j(Q_G(x)), \quad G_j(x) = D_j(x) \cap (-\infty, U_j(x)], \quad G(x) = \prod_j G_j(x), \tag{17}$$

$$C_{\text{gwc}}(x) = \{y : \mathbf{E}(x,y) \in G(x)\}. \tag{18}$$

For a finite lower endpoint a_j, (14) uses $F_j(t, a_j)$, $-1/\sqrt{N}$, and the critical maximum when $t > m_j$ and $z^*(t) \geq a_j$. For $a_j = -\infty$ it simplifies to

$$\psi_{j,\mathbb{R}}(t) = \begin{cases} \sqrt{(t-m_j)^2/s_j^2 + 1/N}, & t > m_j, \\ 1/\sqrt{N}, & t \leq m_j. \end{cases}$$

Proposition D.1 (GWC containment). *For every realized calibration sample and test input,*

$$C_{\text{full}}(x) \subseteq C_{\text{gwc}}(x).$$

Proof. For a candidate in $D(x)$, $\Phi_{\mathbf{e}}(\mathbf{E}^i) \leq \Phi_G(\mathbf{E}^i; x)$ for every i, whence $Q_{\mathbf{e}} \leq Q_G$. Full acceptance implies $g_j(e_j) \leq Q_{\mathbf{e}} \leq Q_G$, and Lemma B.1 gives $e_j \leq U_j$. □

The dependence of $D(x)$ on the test input causes no difficulty: this is a pathwise containment argument, not a claim that GWC calibration scores formed using that input are exchangeable on their own.

E From full cells to surface envelopes

Assume $G(x)$ is nonempty. For each coordinate, partition G_j using its two endpoints and the distinct calibration residuals strictly between them:

$$a_j = \eta_{j,0} < \eta_{j,1} < \cdots < \eta_{j,K_j} = U_j, \qquad I_{j,k} = [\eta_{j,k-1}, \eta_{j,k}] \cap \mathbb{R}.$$

There are at most $n+1$ cells. We deliberately use closed cells, a cover with shared endpoints. This prevents boundary exclusions when residuals have atoms. A singleton G_j is represented by one singleton cell. Distinct interior knots avoid zero-width duplicate cells. The proofs allow any finite ordered interval cover, though Section G.1 uses the calibration partition.

A full cell $\mathcal{I}_h = \prod_j I_{j,h_j}$ has exact local score

$$\Phi_h^{\text{exact}}(\mathbf{t}) = \sup_{\mathbf{z}\in\mathcal{I}_h} \Phi_{\mathbf{z}}(\mathbf{t}) = \max_j \psi_{j,I_{j,h_j}}(t_j).$$

Enumerating all h costs up to $(n+1)^d$ cells. Instead fix only the jth coordinate cell and let the others range over GWC:

$$H_j(\mathbf{t}) = \max_{\ell\neq j} \psi_{\ell,G_\ell}(t_\ell), \tag{19}$$

$$\overline{\Phi}_{j,k}(\mathbf{t}) = \max\{\psi_{j,I_{j,k}}(t_j), H_j(\mathbf{t})\} = \sup_{\substack{z_j\in I_{j,k}\\ z_\ell\in G_\ell,\ \ell\neq j}} \Phi_{\mathbf{z}}(\mathbf{t}), \tag{20}$$

$$\overline{Q}_{j,k} = \mathcal{Q}_\alpha(\overline{\Phi}_{j,k}(\mathbf{E}^1), \ldots, \overline{\Phi}_{j,k}(\mathbf{E}^n)). \tag{21}$$

Set $H_1 = -\infty$ for $d = 1$. Every transformation in these formulas is given explicitly by (14). For $h_j = k$ we have the pointwise chain

$$\Phi_{\mathbf{e}} \le \Phi_h^{\text{exact}} \le \overline{\Phi}_{j,k} \le \Phi_G \quad \text{whenever } \mathbf{e}\in\mathcal{I}_h. \tag{22}$$

The envelope enlarges the candidate domain before taking the supremum. In general a quantile of pointwise maxima is not the maximum of the corresponding quantiles. Thus no equality with exact full-cell LWC is asserted.

Define the accepted coordinate piece and its upper endpoint by

$$A_{j,k} = I_{j,k} \cap (-\infty, \omega_j(\overline{Q}_{j,k})], \tag{23}$$

$$B_{j,k} = \begin{cases} \min\{\eta_{j,k}, \omega_j(\overline{Q}_{j,k})\}, & A_{j,k} \neq \varnothing, \\ -\infty, & A_{j,k} = \varnothing. \end{cases} \tag{24}$$

For a finite lower endpoint, nonemptiness is exactly $\omega_j(\overline{Q}_{j,k}) \ge \eta_{j,k-1}$; a value $-\infty$ never represents a finite residual. In particular, equality retains the singleton at the cell's left endpoint. Let

$$L_j = \max_{1\le k\le K_j} B_{j,k}, \qquad R_{\text{env}}(x) = D(x) \cap \prod_j (-\infty, L_j], \qquad C_{\text{env}}(x) = \{y : \mathbf{E}(x,y) \in R_{\text{env}}(x)\}. \tag{25}$$

If any coordinate has no accepted piece, return the empty set. Otherwise R_{env} is the coordinatewise lower enclosure of the pieces; it may fill gaps between them. It is not being identified with the exact local union.

Theorem E.1 (Signed envelope validity). *Pathwise, $C_{\text{full}}(x) \subseteq C_{\text{env}}(x) \subseteq C_{\text{gwc}}(x)$. Consequently $\Pr\{Y^{n+1} \in C_{\text{env}}(X^{n+1})\} \ge 1-\alpha$.*

Proof. Take $y \in C_{\mathrm{full}}(x)$. Proposition D.1 implies that its residual $\mathbf{e}$ belongs to $G(x)$. For each j, choose a covering cell $I_{j,k}$ containing e_j. Every calibration observation satisfies $\Phi_{\mathbf{e}}(\mathbf{E}^i) \le \overline{\Phi}_{j,k}(\mathbf{E}^i)$. Quantile monotonicity and full acceptance therefore yield

$$g_j(e_j) \le \max_\ell g_\ell(e_\ell) \le Q_{\mathbf{e}} \le \overline{Q}_{j,k}, \quad e_j \le \omega_j(\overline{Q}_{j,k}), \quad e_j \in A_{j,k}, \quad e_j \le B_{j,k} \le L_j.$$

This proves the first containment simultaneously for all coordinates without a Bonferroni correction. Each nonempty $A_{j,k}$ lies in G_j, so $L_j \le U_j$, which proves the second. The coverage statement follows from Theorem C.1. □

F Why it improves the old separated shortcut

Here the comparison uses the *same nonnegative calibration residuals*, test score function, level α, and calibration partition, with $D_j = [0, \infty)$. It does not compare signed residuals to capped or shifted residuals. Assume $s_j > 0$.

The mathematical old GWC transformation is the same exact supremum $\max_j \psi_{j,[0,\infty)}(t_j)$ as (15). Its implementation evaluates the endpoint at zero, the limit at infinity, and the stationary point. For $t_j > m_j$, an admissible stationary point is exactly the maximum in (13); for $t_j < m_j$ the stationary point is a minimum and cannot exceed both endpoints. Thus, in exact arithmetic and without jitter, $G^{\mathrm{old}} = G$ for the same nonnegative scores and rank. The implementation's nonnegative jitter can only enlarge the GWC quantile; the theoretical comparison below is made with the unperturbed closed rules. Define the old GWC box G^{old} and

$$c_\ell = \inf_{z \in G_\ell^{\mathrm{old}}} \frac{\mu_\ell(z)}{\sigma_\ell(z)}, \qquad r_\ell(h) = \inf_{z \in I_{\ell,h}^{\mathrm{old}}} \sigma_\ell(z). \tag{26}$$

For nonnegative residuals $m_\ell \ge 0$ and $G_\ell^{\mathrm{old}} = [0, U_\ell^{\mathrm{old}}]$. Differentiating $\mu(z)/\sigma(z)$ gives

$$\frac{d}{dz}\frac{\mu(z)}{\sigma(z)} = \frac{s^2 - m(z-m)}{N\sigma(z)^3}.$$

Its interior critical point, when present, is a maximum. Thus

$$c_\ell = \min\left\{ \frac{\mu_\ell(0)}{\sigma_\ell(0)}, \frac{\mu_\ell(U_\ell^{\mathrm{old}})}{\sigma_\ell(U_\ell^{\mathrm{old}})} \right\},$$

with the second expression interpreted as $1/\sqrt{N}$ at infinity, and

$$r_\ell(h) = \sqrt{s_\ell^2 + \mathrm{dist}(m_\ell, I_{\ell,h}^{\mathrm{old}})^2/N}.$$

The old separated score on a full cell is

$$\Phi_h^{\mathrm{sep}}(\mathbf{t}) = \max_\ell \{t_\ell / r_\ell(h_\ell) - c_\ell\}, \qquad \mathbf{t} \ge 0.$$

Let $p_\ell = \mathrm{proj}_{G_\ell^{\mathrm{old}}}(m_\ell)$ and $\bar{s}_\ell = \sigma_\ell(p_\ell) = \inf_{z \in G_\ell^{\mathrm{old}}} \sigma_\ell(z)$. If $m_\ell \in G_\ell^{\mathrm{old}}$, then $p_\ell = m_\ell$ and $\bar{s}_\ell = s_\ell$. Otherwise $\bar{s}_\ell$ is the minimum scale on the clipped domain, which need not equal s_ℓ. Choose each off-coordinate old cell to contain p_ℓ. Whenever the old mean cell intersects GWC it has this property after clipping. Its minimum scale is therefore $\bar{s}_\ell$. The old row score is

$$\Phi_{j,k}^{\mathrm{old}}(\mathbf{t}) = \max\left\{ \frac{t_j}{r_j(k)} - c_j, \max_{\ell \ne j}\left(\frac{t_\ell}{\bar{s}_\ell} - c_\ell\right) \right\}. \tag{27}$$

Its quantile and clipped coordinate pieces define L_j^{old} as in (21)–(25), using the old cells. If the old mean cell is unavailable, the manuscript shortcut returns its GWC box.

Lemma F.1 (Pointwise score domination). *For* $\mathbf{t} \geq 0$ *and the same GWC box and cells,* $\overline{\Phi}_{j,k}(\mathbf{t}) \leq \Phi^{\mathrm{old}}_{j,k}(\mathbf{t})$.

Proof. For $z \in I_{j,k}$, $\sigma_j(z) \geq r_j(k) > 0$ and $\mu_j(z)/\sigma_j(z) \geq c_j$. Nonnegativity gives

$$F_j(t_j, z) = \frac{t_j}{\sigma_j(z)} - \frac{\mu_j(z)}{\sigma_j(z)} \leq \frac{t_j}{r_j(k)} - c_j.$$

Taking the supremum proves the active-coordinate inequality. For $\ell \neq j$ and $z \in G_\ell$, $\sigma_\ell(z) \geq \bar{s}_\ell$, so similarly $F_\ell(t_\ell, z) \leq t_\ell/\bar{s}_\ell - c_\ell$. Take the off-coordinate suprema and then the maximum. Both steps explicitly use $t_\ell \geq 0$; reversing its sign reverses the comparison of the scale terms. □

Theorem F.2 (Uniform weak improvement over the old shortcut). *Under the preceding conditions, with closed-cell and weak-acceptance conventions for both constructions,*

$$C_{\mathrm{env}}(x) \subseteq C_{\mathrm{old}}(x), \qquad L_j \leq L^{\mathrm{old}}_j \quad (j \leq d)$$

whenever both coordinate bounds represent nonempty sets. The set containment also holds when the envelope is empty or when the old shortcut falls back to GWC. In particular, each outcome interval and the resulting rectangular volume are weakly smaller, while finite-sample coverage remains at least $1 - \alpha$.

Proof. Lemma F.1 and quantile monotonicity give $\overline{Q}_{j,k} \leq Q^{\mathrm{old}}_{j,k}$. The signed link is increasing, so each accepted envelope piece is contained in its corresponding old piece. Taking coordinatewise suprema yields $L_j \leq L^{\mathrm{old}}_j$. Taking score sublevel sets preserves containment in outcome space. On an old GWC fallback sample, Theorem E.1 gives the result directly. □

The proof also works if the envelope's GWC box is a subset of the old box and its cells refine the old cells: each supremum is then over a smaller domain. Thus a harmless outward rounding in the old GWC computation does not reverse the mathematical comparison.

What the statement does and does not compare. The old manuscript uses some strict cell tests and assumes distinct residuals; the present note specifies a closed version to cover atoms as well. A literal comparison to code with a different tie policy requires numerical checking or harmonizing those policies; it is not justified by ignoring endpoints. The repository's old implementation also adds positive numerical jitter. The mathematical theorem is about the explicitly defined rules, not arbitrary floating-point perturbations. The paired experiment audit separately compares the actual implementations. During that audit, capped CQR scores exposed a specific bug in the former implementation: `argmax` selected the first tied zero below the mean, which could select a $[0, 0]$ cell not containing the strictly positive mean. The off-coordinate scale then exceeded s_ℓ, so (27) no longer described that code. Selecting the last value at or below the mean (right-sided insertion rank) restores a mean-containing cell. All six saved failing examples satisfy containment after this correction; additional atom-heavy regression tests cover the same issue. Thus the theorem compares the mathematical old shortcut, including this necessary tie fix, and does not claim dominance over the buggy historical executable. The theorem is a containment result, not a claim that the envelope has higher coverage than a larger old set. Strict size improvement need not occur. Exact full-ratio cell LWC can be tighter than the surface envelope, and CQHR has a different scalarization; neither is dominated by this argument.

G Computation and a justified search rule

Compute m, s and the GWC scores in $O(nd)$ operations. Sort the distinct calibration coordinates in $O(dn \log n)$. Precompute $V_{i\ell} = \psi_{\ell,G_\ell}(E^i_\ell)$ and all $H_{ij} = \max_{\ell \neq j} V_{i\ell}$ in $O(nd)$ using left and right cumulative maxima. Each surface quantile then requires $O(n)$ arithmetic and an order statistic selection, with no factor of d inside the surface search.

Lemma G.1 (Rightmost surviving cell)**.** *For any ordered interval cover, the largest $B_{j,k}$ is the endpoint of the rightmost nonempty $A_{j,k}$. Thus scanning from K_j downwards and stopping at the first nonempty piece computes L_j exactly, without a monotone predicate.*

Proof. If cell k survives and $h < k$, then $B_{j,h} \le \eta_{j,h} \le \eta_{j,k-1} \le B_{j,k}$ whenever h survives. An empty piece cannot increase the maximum. All cells to the right have already been checked and are empty. $\square$

The worst-case cost is $O(dn^2 + dn \log n)$, and the actual search cost is $O(n \sum_j M_j)$ when M_j cells are inspected. The implementation records these counts. This proof does not claim that the values $B_{j,k}$ are monotone: they drop to the empty sentinel after a failure.

G.1 A range where binary search is proved

For $n > 1$ put $b_j = m_j - s_j/\sqrt{n-1}$. We prove a stronger search fact than the usual restriction to cells above the empirical mean, but make no unconditional prefix claim for the cells below b_j.

Lemma G.2 (Rank-count predicate above b_j)**.** *Consider a calibration-partition cell $I_{j,k} = [\ell, u]$ with finite $\ell > b_j$ and $u > \ell$. Set $h_i = H_j(\mathbf{E}^i)$. Then*

$$A_{j,k} \neq \varnothing \iff \#\{i : E_j^i < \ell,\ h_i < g_j(\ell)\} < r. \tag{28}$$

Consequently the survival predicate is a true prefix followed by a false suffix on these cells, and binary search is valid on this part of the partition.

Proof. First $g_j(\ell) > -1/\sqrt{N}$ is equivalent to $\ell > b_j$ by (7). For fixed $t = \ell$, (12) shows that $F_j(\ell, z)$ has either no interior maximum or is decreasing for $z \ge \ell$. Its supremum over $[\ell, \infty)$ is the larger of $g_j(\ell)$ and $-1/\sqrt{N}$, hence is $g_j(\ell)$. More explicitly, if $\ell < m_j$ the interior critical point is a minimum; if $\ell \ge m_j$ the derivative is negative on this half-line.

If $t < \ell$, then $F_j(t, z) < F_j(\ell, z)$ at every finite z, and its limit at infinity is strictly below $g_j(\ell)$. Continuity, including that limit, therefore implies $\psi_{j,I_{j,k}}(t) < g_j(\ell)$. If $t = \ell$, the supremum equals $g_j(\ell)$. If $t > \ell$, evaluating at $z = \ell$ already gives a value strictly larger than $g_j(\ell)$. It follows that the ith surface score is strictly below $g_j(\ell)$ exactly when both inequalities inside the count in (28) hold. Its rth order statistic is at least $g_j(\ell)$ exactly when fewer than r scores are strictly below this value. Link inversion proves the equivalence. As ℓ increases, $g_j(\ell)$ increases, so the counted sets are nested and their cardinalities cannot decrease. $\square$

In fact, the proof of the rank predicate uses only $z \ge \ell$ and the left endpoint; no sign assumption is placed on the calibration residuals. One can check the top cell first, use binary search on the cells with left endpoint above b_j, and scan backwards below that range only if no upper cell survives. This has $O(dn \log n)$ search cost when every coordinate boundary is found in the proved range, with the unconditional backward bound otherwise. Observed prefix behavior outside the proved range must not be substituted for a proof. The initial reference implementation uses unconditional backward search so its correctness does not depend on this optional acceleration.

G.2 Direct order-statistic localization

Binary search is not necessary in the certified range. For each coordinate set

$$W_{ij} = \max\{E_j^i,\ \omega_j(H_j(\mathbf{E}^i))\}, \qquad T_j = W_{(r),j},$$

where $W_{(r),j}$ is the rth order statistic, with the same infinite-rank convention. The inverse is evaluated separately at each off-coordinate score; it is not the inverse of an aggregate quantile.

Proposition G.3 (One-order-statistic localization)**.** *For a nondegenerate cell with left endpoint $\ell > b_j$,*

$$A_{j,k} \neq \varnothing \iff \ell \leq T_j.$$

In particular every cell with left endpoint exceeding $\max\{b_j, T_j\}$ *can be discarded without evaluating a surface quantile.*

Proof. The extended inverse convention gives $h_i < g_j(\ell) \iff \omega_j(h_i) < \ell$ even when h_i is outside the open range of g_j. Hence

$$\{E_j^i < \ell,\ h_i < g_j(\ell)\} = \{\max(E_j^i, \omega_j(h_i)) < \ell\} = \{W_{ij} < \ell\}.$$

There are fewer than r such values exactly when $W_{(r),j} \geq \ell$. Apply Lemma G.2. The discarded cells are all in that lemma's range, so this pruning makes no claim about the remaining lower cells. ☐

Select T_j in $O(n)$ time and find the last partition left endpoint at or below $\max\{b_j, T_j\}$ by insertion search in $O(\log n)$ time. Start the unconditional backward scan there. If this cell has left endpoint above b_j, it is the rightmost survivor and only its surface quantile is needed to compute the actual clipped endpoint $B_{j,k}$. The endpoint is generally *not* T_j: the order statistic localizes the cell, after which (21)–(25) still determine its boundary. If the candidate cell is below the certified range, continue backward normally. Thus search costs $O(nd + d\log n)$ when all coordinate boundaries lie in the certified range, after sorting and off-coordinate preprocessing, with the unconditional worst-case bound otherwise. The optional implementation `search="rank"` uses this pruning and an outward numerical guard near ties. It need not be faster on typical samples: a backward scan that already checks one cell avoids the extra transformed order statistic.

H Outcome intervals, signed improvements, and edge cases

For absolute residuals $E_j(x, y) = |y - \hat{f}_j(x)|$, $a_j = 0$ and the prediction interval is $[\hat{f}_j(x) - L_j, \hat{f}_j(x) + L_j]$. For ordered fitted quantiles $q_j^-(x) \leq q_j^+(x)$, signed CQR scores satisfy

$$\begin{aligned} E_j(x, y) &= \max\{q_j^-(x) - y,\ y - q_j^+(x)\} \\ &= \left| y - \frac{q_j^-(x) + q_j^+(x)}{2} \right| - \frac{q_j^+(x) - q_j^-(x)}{2}, \qquad a_j(x) = -\frac{q_j^+(x) - q_j^-(x)}{2}. \end{aligned}$$

Consequently

$$E_j(x, y) \leq L_j \iff q_j^-(x) - L_j \leq y \leq q_j^+(x) + L_j, \qquad \text{length}_j = (q_j^+ - q_j^-) + 2L_j.$$

A negative L_j shrinks the base interval. A bound below a_j means an empty interval, not a singleton created by clamping. A bound equal to a_j gives the singleton at the midpoint. Mixed signs across coordinates are allowed; GWC inflation in one or all coordinates does not rule out local shrinkage. The envelope's computations for one coordinate still depend on the other coordinates through H_j, so a general sign-reflection shortcut is not implied.

Small calibration samples. If $r = N$, all conformal quantiles are $+\infty$ and the procedure returns the entire attainable domain. An implementation must preserve this case, not replace the quantile by the largest observed score.

Zero calibration variance. To extend the definition to all finite calibration samples, define the full conformal reference scale to be the augmented sample standard deviation when positive, and 1 when the entire augmented coordinate is constant. This rule is symmetric, so Theorem C.1 still holds. If any observed $s_j = 0$, the reference implementation returns the whole attainable domain, a valid superset of that same reference set. Otherwise all proofs above apply. This conservative fallback avoids division by zero; it can be costly for heavily capped scores. The positive-score dominance theorem excludes these samples because the unregularized old shortcut also lacks a positive scale. A practical regularized variant could instead add a predetermined positive ridge to every augmented variance symmetrically, but its comparison must then be made to an old shortcut with matching regularization.

Infinity, empty boxes, and ties. If GWC is empty, return empty immediately. Infinite cell endpoints use the limits in (14). An infinite upper coordinate makes volume infinite unless some coordinate interval is empty or has zero length, in which case rectangular Lebesgue volume is zero. Closed cells and weak inequalities are maintained across calibration ties and candidate ties. The finite-sample coverage guarantee is marginal over the calibration/test experiment; it is not a guarantee conditional on the realized calibration sample or on every x.

References

[1] G. Shafer and V. Vovk (2008). A Tutorial on Conformal Prediction. *Journal of Machine Learning Research* 9:371–421. `https://jmlr.org/papers/v9/shafer08a.html`.

[2] Y. Romano, E. Patterson, and E. J. Candès (2019). Conformalized Quantile Regression. *Advances in Neural Information Processing Systems* 32. `https://proceedings.neurips.cc/paper/2019/hash/5103c3584b063c431bd1268e9b5e76fb-Abstract.html`.